

TRAWSGOED, United Kingdom

WMO: 035030

Lat: 52.3442N

Lon: 3.9481W

Elev: 203

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.6	26.4	21.1	16.0	29.3	23.8	18.1	31.3	25.5	41.0	23.4	43.4	1.1	130	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	75.6	64.5	72.0	62.8	69.1	61.4	66.0	72.4	64.3	69.6	62.8	67.3	8.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.7	88.9	68.2	62.1	84.1	66.4	60.8	80.2	65.1	30.8	72.7	29.5	69.7	28.4	67.4	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	18.5	16.4	DB	19.2	82.6	3.8	3.6	16.4	85.2	14.2	87.3	12.1	89.3	9.3	91.9	(4)
(5)			WB	19.7	70.6	3.4	3.5	17.3	73.1	15.3	75.1	13.4	77.1	10.9	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	41.6	41.7	43.6	46.9	51.9	56.7	59.7	59.2	56.1	51.9	46.1	42.5		
	DBStd	8.48	6.01	5.71	5.43	5.21	5.10	4.60	4.00	3.90	4.38	5.31	6.04	6.82		
	HDD50	1292	263	236	207	118	36	3	0	0	4	42	142	241		
	HDD65	5541	726	652	664	542	407	252	174	185	267	407	567	698		
	CDD50	1244	3	3	8	26	94	205	299	284	189	100	25	8		
	CDD65	19	0	0	0	0	1	4	8	4	2	0	0	0		
	CDH74	179	0	0	0	1	8	55	87	20	7	1	0	0		
	CDH80	23	0	0	0	0	0	8	13	2	0	0	0	0		
	Wind	WSAvg	6.6	7.8	7.8	7.5	6.8	6.7	5.8	5.5	5.4	6.4	6.4	7.4		
	Precipitation	PrecAvg	46.2	4.7	3.3	3.6	2.9	2.8	3.0	2.9	3.7	4.1	4.9	5.4		
		PrecMax	56.8	8.7	8.1	9.1	5.4	6.2	6.3	5.8	6.9	8.5	10.7	9.3		
		PrecMin	35.2	0.2	0.0	1.3	0.5	0.9	0.6	0.5	0.1	0.6	1.7	1.8		
		PrecStd	5.7	2.2	1.8	1.8	1.5	1.4	1.6	1.3	1.5	1.9	2.2	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	58.7	62.5	70.0	74.5	80.8	81.7	76.5	74.7	67.9	60.1	56.7		
		MCWB	52.9	50.0	52.2	56.8	62.3	65.6	66.4	66.2	64.7	60.5	57.2	54.3		
	2%	DB	53.3	53.3	57.3	63.0	69.1	73.7	76.0	71.9	69.8	63.4	58.1	55.3		
		MCWB	51.1	50.2	50.7	53.7	59.3	63.1	65.0	63.8	62.3	58.9	55.7	53.3		
	5%	DB	51.6	51.6	54.1	58.8	64.9	69.2	71.9	68.8	66.6	61.5	56.5	53.7		
		MCWB	49.3	49.0	49.9	51.8	57.3	60.7	63.2	62.2	60.6	57.9	54.2	51.6		
	10%	DB	50.2	50.1	52.0	55.7	61.4	66.1	68.3	66.5	64.3	59.8	54.8	52.0		
		MCWB	48.0	47.6	49.0	50.6	54.7	59.2	61.7	61.3	59.3	56.6	52.5	50.0		
	0.4%	WB	53.4	53.0	54.4	58.0	64.2	67.7	69.6	67.4	66.4	62.3	58.2	54.9		
		MCDB	54.8	55.4	58.9	67.8	72.1	77.2	77.6	74.0	71.7	64.8	59.7	55.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.7	51.0	52.1	54.8	60.6	63.8	66.2	65.2	63.7	60.2	56.2	53.6		
		MCDB	53.3	52.8	55.2	60.7	67.3	71.7	73.7	69.6	67.6	62.6	57.6	55.1		
	5%	WB	49.9	49.5	50.8	53.0	58.1	61.7	64.1	63.7	61.8	58.6	54.6	52.0		
		MCDB	51.3	51.3	53.5	57.5	63.5	67.8	70.1	67.3	65.2	60.8	56.1	53.3		
	10%	WB	48.3	47.9	49.3	51.3	55.9	60.0	62.4	62.3	60.2	57.0	52.7	50.2		
		MCDB	49.9	49.9	51.8	55.2	59.9	65.0	67.1	65.6	63.4	59.4	54.6	51.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	11.0	13.1	15.2	15.2	15.1	14.2	13.7	14.7	12.4	11.4	10.4		
		MCDBR	9.7	11.2	15.9	20.1	21.3	22.2	22.5	17.3	19.0	13.3	10.2	9.4		
		MCWBR	9.2	9.1	11.2	13.0	13.3	13.5	13.2	10.9	12.2	9.8	8.9	9.2		
		MCDBR	9.5	10.0	13.7	17.5	18.7	19.3	18.5	14.7	15.7	11.8	9.5	9.1		
	5% WB	MCWBR	9.5	9.0	10.8	11.9	12.6	12.4	12.0	10.1	11.2	9.7	9.0	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.331	0.344	0.374	0.395	0.401	0.391	0.392	0.375	0.367	0.353	0.329	0.321		
	taud		2.292	2.337	2.268	2.240	2.269	2.327	2.332	2.393	2.408	2.402	2.354	2.272		
	Ebn at Noon		205	241	256	266	270	273	271	270	258	235	205	185		
	Edh at Noon		21	26	34	39	40	38	37	33	29	24	19	18		
(44) All-Sky Solar Radiation	RadAvg		229	448	820	1278	1588	1711	1571	1302	968	541	277	173		
	RadStd		22	46	82	120	98	167	156	95	70	48	27	20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page